AI / ML

Transformers & Attention

The architecture behind GPT, BERT, and modern AI. 'Attention is all you need' — explained visually.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

The Problem Attention Solves

Old RNNs read word-by-word and forgot the beginning. Attention looks at everything at once.

What does 'it' refer to? → the ANIMAL (not the street).

Attention lets 'it' directly query every other word — no forgetting.

Swap 'tired' → 'wide' and 'it' now means the STREET. Attention resolves this dynamically per context.

STEP 2/6

Q, K, V: Queries, Keys, Values

Borrowed from databases: query the keys, get weighted values.

Query (Q)
'what am I looking for?'

Key (K)
'what do I contain?'

Value (V)
'what do I actually say?'

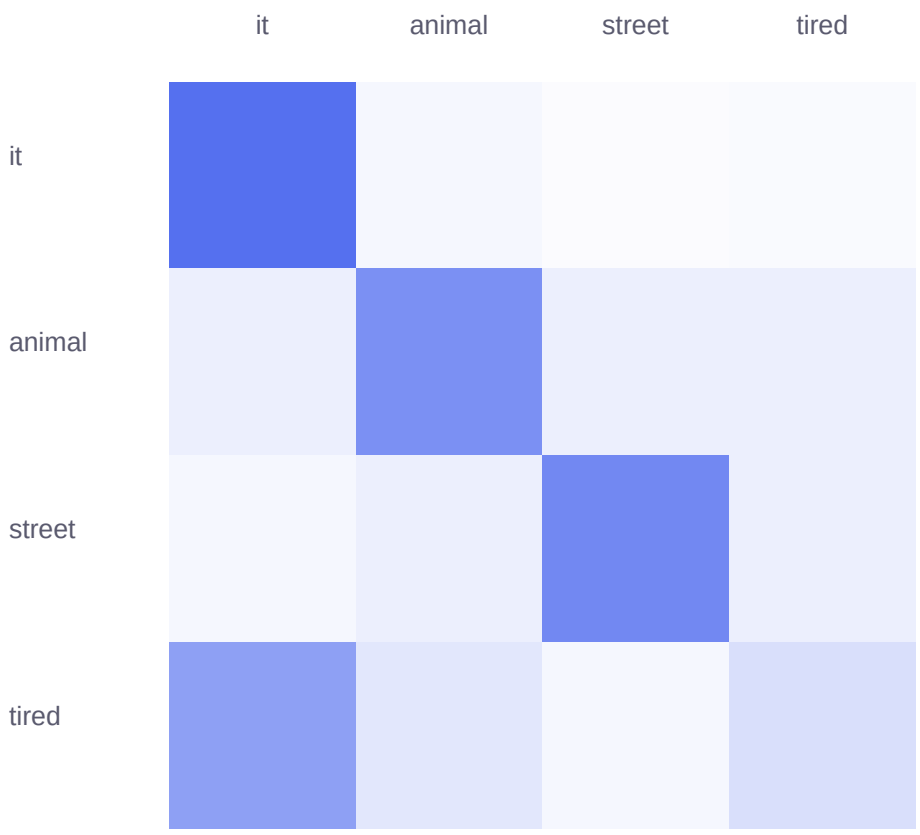
$$\text{Attention}(Q,K,V) = \text{softmax}(QK^T/\sqrt{d}) \cdot V$$

- Score = how well each query matches each key (dot product).
- Softmax → weights summing to 1. Output = weighted mix of values.
- √d scaling keeps gradients stable as dimensions grow.

STEP 3/6

Attention Heatmap

Darker = stronger attention. See what the model 'looks at.'

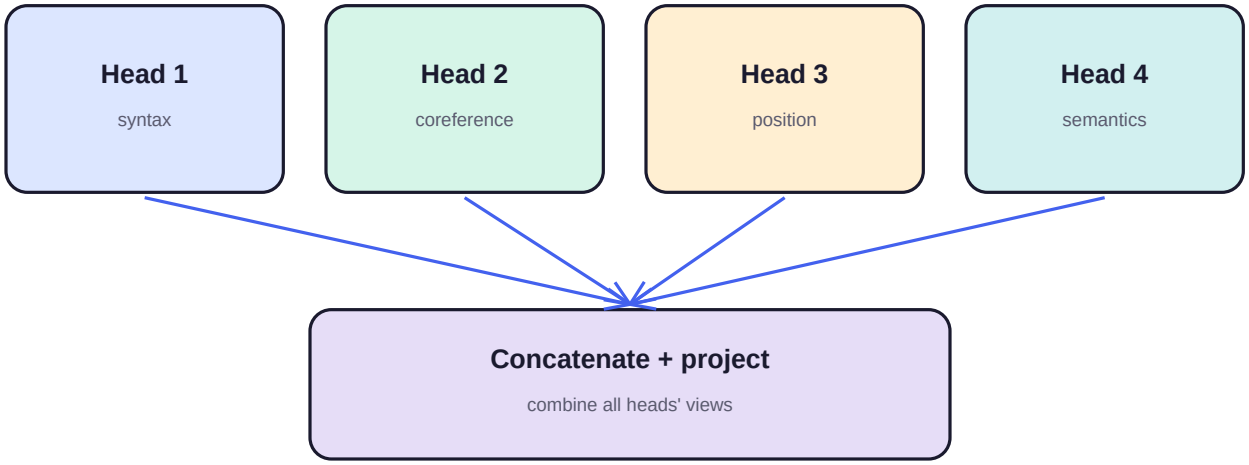


Row 'it' attends mostly to 'animal' (0.9) — the model resolved the pronoun correctly.

STEP 4/6

Multi-Head Attention

Not one attention — many, in parallel, each learning different relations.

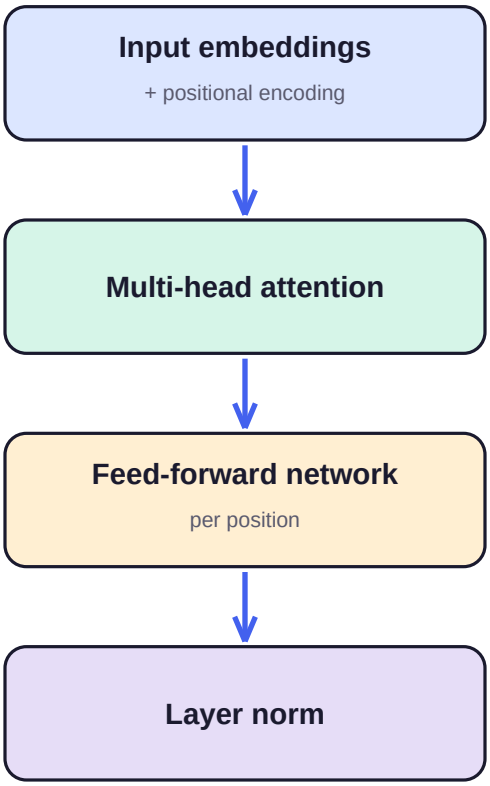


Each head learns a different lens; together they see the whole picture.

STEP 5/6

The Full Block: Attention + Feed-Forward

Repeated N times (GPT-3: 96×). Residuals + normalization keep it trainable.



Residual connections (x + sublayer(x)) let gradients flow through 100+ layers.

Positional encoding injects order — attention itself is order-blind.

STEP 6/6

Why Transformers Won

Parallelism changed everything.

Parallel training

RNNs: sequential. Transformers: all positions at once → GPUs fully fed.

Long context

Attention reaches across thousands of tokens (and growing).

Scale laws

Bigger models + more data + more compute = predictably better. No plateau yet.

Transfer learning

Pre-train once on the internet; fine-tune for anything.

STEP ✓/✓

Key Takeaways

Transformers — Remember

- Attention(Q,K,V) = softmax(QK■/√d)·V — queries match keys, values get mixed by weight.
- Multi-head = parallel attentions, each learning different linguistic relations.
- Blocks stack attention + feed-forward with residuals and layer norm, repeated N×.
- Parallelism over sequence positions is why transformers train at supercomputer scale.
- Pre-train broadly, fine-tune narrowly — the modern recipe.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev